

Trainings ▾

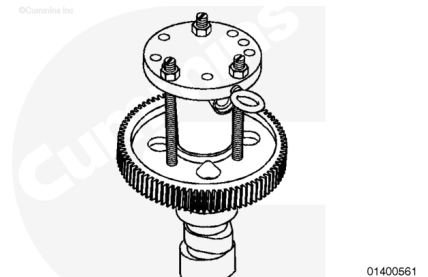
Remove

Use the Camshaft Gear Puller Kits, Part Numbers 3164375, and 3164925, to remove camshaft gear.

Note : Both kit part numbers are necessary when removing the camshaft gear.

Attach the puller to the camshaft gear. Remove the gear.

Remove the puller.

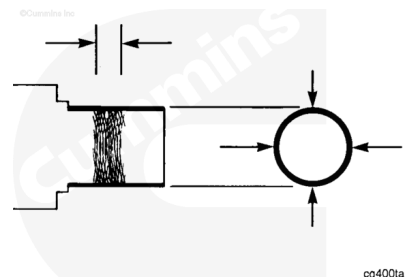


Inspect for Reuse

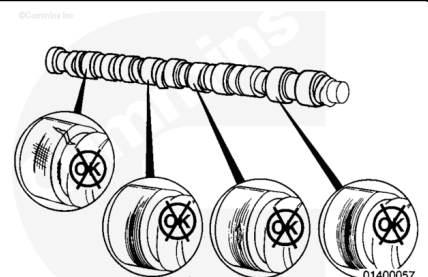
Inspect the camshaft and camshaft gear for reuse. Check for fretting damage. The camshaft **must** be replaced if the fretting damage is more than 3 mm [1/8 in] wide.

Measure the outside diameter of the camshaft.

Camshaft Outside Diameter (Gear Location)		
mm		in
80.10	MIN	3.154
80.12	MAX	3.154

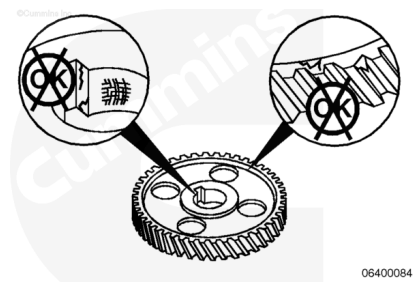


Inspect the camshaft. Refer to Procedure 001-008 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-008.html>).



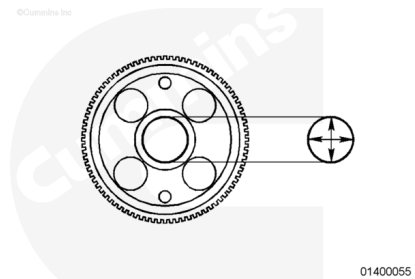
Inspect the camshaft gear for cracks or chipped or broken teeth.

Inspect the bore of the gear for fretting or burrs. Refer to Procedure 001-012 in Section 1 (/qs3/pubsys2/xml/en/procedures/89/89-001-012.html).



Measure the inside diameter of the camshaft gear.

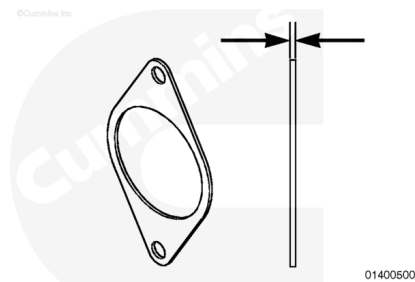
Camshaft Gear Inside Diameter		
mm		in
80.00	MIN	3.150
80.02	MAX	3.150



The gear **must** be replaced if the inside diameter is **not** within specification.

Inspect the thrust bearing for damage.

Measure the thickness. Refer to Procedure 001-056 in Section 1 (/qs3/pubsys2/xml/en/procedures/89/89-001-056.html).



Install

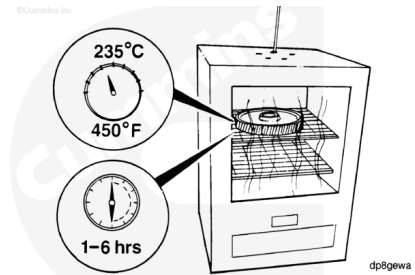
⚠ WARNING ⚠

To reduce the possibility of severe burns, wear protective gloves when installing the heated ring gear.

⚠ CAUTION ⚠

Do not exceed the specified time or the temperature. Damage to the gear and the gear teeth will result.

⚠ CAUTION ⚠



Do not attempt to install the gear without using heat. Damage to the camshaft and camshaft gear can result.

Use an oven with the temperature adjusted to 235°C [450°F]. Heat the gear in the oven for a minimum of one hour but **not** more than a maximum of six hours. The inside diameter of the gear will become larger and will simplify the installation.

⚠ CAUTION ⚠

Allow the gear to air cool. Do not use water or oil to reduce the cooling time. It will cause the gear to crack.

Install the thrust bearing onto the camshaft.

If removed, install the camshaft key into the slot on the camshaft.

Note : There is **only** a straight key option. It is **not** necessary to check/adjust static injection timing on the QSK23 engine.

The timing mark on the camshaft gear **must** be visible from the front of the gear after it is installed on the camshaft.

Remove the gear from the oven and install it on the camshaft. The keyway in the gear **must** be aligned with the key in the camshaft.

